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(17)

Control #1

$$\begin{cases} 2x + 2y - 4z = 9 \\ -x + 2y + 4z = -17 \\ 2x - 3y + 4z = 1 \end{cases}$$

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$$\left(\begin{array}{ccc|c} 2 & 2 & -4 & 9 \\ -1 & 2 & 4 & -17 \\ 2 & -3 & 4 & 1 \end{array}\right) \begin{array}{l} f_2 \to -\left(\frac{1}{2}\right) f_1 + f_2 \\ f_3 \to -\left(\frac{2}{2}\right) f_1 + f_3 \end{array}$$

$$\left(\begin{array}{ccc|c} 2 & 2 & -4 & 9 \\ 0 & 3 & 2 & -\frac{25}{2} \\ 0 & -5 & 8 & -8 \end{array}\right) f_3 \to -\left(\frac{-5}{3}\right) f_2 + f_3$$

$$\left(\begin{array}{ccc|c} 2 & 2 & -4 & 9 \\ 0 & 3 & 2 & -\frac{25}{2} \\ 0 & 0 & \frac{34}{3} & -\frac{173}{6} \end{array}\right)$$

$$0x + 0y + \frac{34}{3}z = -\frac{173}{6}$$

$$z = \frac{-173(3)}{6*34}$$

$$\boxed{z = -2.54}$$

$$0x + 3y + 2(-2.54) = -\frac{25}{2}$$

$$3y - 5.08 = -\frac{25}{2}$$

$$3y = -\frac{25}{2} + 5.08$$

$$3y = -7.42$$

$$y = \frac{-7.42}{3}$$

$$\boxed{y = -2.47}$$

$$2x + 2(-2.47) - 4(-2.54) = 9$$

$$2x - 4.94 + 10.16 = 9$$

$$2x = 9 - 10.16 + 4.94$$

$$2x = 3.78$$

$$x = \frac{3.78}{2} \Rightarrow \boxed{x = 1.89}$$

$$\begin{cases} x = 1.89 \\ y = -2.47 \\ z = -2.54 \end{cases}$$

Escaneado con CamScanner